

Group Assignment/Project

Model Solar Powered Car

The model solar car comprises a 10W solar panel mounted on a model car chassis. Also mounted on the chassis are a dc motor, gearbox, steering servo, and radio control module. **ONLY** the servo and radio control module may be powered from an auxiliary battery source. You are not allowed to alter any mechanical characteristics of the car. The car requires design and build of a closed-loop feedback controlled buck converter to make the dc motor run from the solar panel. Optional event – car race competition.

Project Description:

1. Complete some introductory lab exercises.
2. Carry out an investigation to determine the solar panel and dc motor characteristics and hence determine a suitable design strategy.
2. Design, simulate, and build a buck converter to interface between the solar panel and the dc motor. A breadboard can be used for initial circuit testing, but the final circuit used for the race must be mounted on veroboard or a printed circuit board.
3. Test your vehicle to determine its performance. The vehicle must be able to start moving under its own power as soon as sunlight falls on the panel. The race will be in a straight line of about 10 metres.

Design constraints/requirements:

1. A buck converter circuit must be used that utilises closed loop feedback control.
2. Control chip must be the TL494.
3. N-channel MOSFET IPP034N03L must be used (and gate **must** be driven with a CMOS inverter constructed from a BS250P PMOSFET and a BS270 NMOSFET).
4. Diode SB240S should be used.
4. Inductor core is of type RM8. You must design and wind the inductor yourselves.
5. Maximum total capacitance in the circuit is 350 μF .

Project Support:

People:	Edsel Villa is based in the Power Electronics lab, Assoc Prof Chris Hann will be covering LTSpice help and assessment, and Teaching Assistants will be available in the Electrical Machines lab, times as stated for labs and tutorials.
Labs:	Three lab exercises that are very useful for this project are the TL494 lab, the buck converter lab and a BJT transistor lab. These will run in the Electrical Machines lab on dates below.
Tutorials:	Drop-in sessions Wednesdays 5-6pm in the Electrical Machines lab (from 12/8). A tutorial will be run to help with the LTSpice assessment (14/8 in A2).
Components:	Semiconductors, inductor core, breadboard, etc. will issued by Edsel Villa on Wednesday 5/8 (5-6pm) in the Machines Lab. Other components are available in the ECE store.
Tools:	Soldering irons and other specialised tools will be provided.
Benches:	Benches with power supplies and oscilloscopes are available in the Electrical Machines lab. Do not spread beyond these benches or take equipment from other benches. The Power Electronics lab will also be available during the mid-semester break.
Test rigs:	Solar panels and dc motors are available for testing in the Electrical Machines Lab.

Practical work (lab investigations, design and testing) is to be carried out in **groups of 3**.

Schedule:

Friday 24 July:	Nominate your own groups on Learn. Three people per group. If you can't find any people to form a group, you must email paul.gaynor@canterbury.ac.nz by this date to be allocated to a group.
Wednesday 5 August (Tutorial time):	Components issued to groups. Go to Machines Lab.
Friday 21 August (or earlier), 5 pm:	LTspice basic circuit simulation submission (5%).
Tuesday 8 September, 5 pm:	Last date for PCB files submitted to Edsel for manufacture.
Wed/Thu/Fri 23/24/25 Sept:	Group Project Inspection (10%).
Mon 12 October, 5 pm:	Group Design Report (15%).
Tue 13 October:	Self and Peer assessment

Introductory lab exercises:

Mon 27 July: 2-4	PWM & TL494 lab
Tue 28 July: 2-4	PWM & TL494 lab
Wed 29 July: 12-2	PWM & TL494 lab
Thu 30 July: 12-2	PWM & TL494 lab
Mon 3 August: 2-4	Buck Converter lab
Tue 4 August: 2-4	Buck Converter lab
Wed 5 August: 12-2	Buck Converter lab
Thu 6 August: 12-2	Buck Converter lab

Don't delay in getting started with working on circuit hardware! Immediately following the components issue date (5/8), you should start working on circuit hardware design and implementation. This work should be carried out concurrently with the simulation development.

Project Inspection:

The **Project Inspection** will take place in the **Power Electronics Lab**. You must **bring a spare copy of both your circuit diagram and your design notes** to the inspection. I will keep your circuit diagram and design notes as a memory aid when assigning marks. A solar panel will be provided as the power source and a solar car motor will be connected as a load. For reliable circuit operation in the project inspection it is **highly recommended** that you have your circuit built on Veroboard (PCB even better). The marking schedule for the inspection will be based around the following sorts of questions and all members of the same group **who attend and participate** will receive the same Project Inspection mark:

- Explain how you achieve the power supply for your PWM chip.
- Demonstrate how the control signal interacts to change the pulse width.
- Demonstrate the gate signal driving your power MOSFET(s).
- Demonstrate how the gate signal changes with mechanical load.
- Demonstrate the current waveform through your inductor.
- Explain the principle of operation of your control system.
- Demonstrate that you can develop torque on a loaded motor

Note: **You must provide a wire loop on your board** so that a current probe can easily be inserted to measure inductor current (approx. 2cm high and 1cm wide).

Your report must be a group report (.pdf file format), which should include all aspects of the design, operation and overall performance of your converter. There will be a late report penalty of 10%, with an additional 10% per day, unless otherwise pre-arranged. This report will be **automatically checked by TurnItIn**.

Use of Generative AI: The use of Artificial Intelligence (AI) tools for the project is summarised in the Table below. Students are always responsible for the accuracy of the submitted works, regardless of which tools are used.

Assessment Item	Permitted use of AI	
Laboratories	Generative AI tools are not restricted for this assessment.	
LT Spice Assignment	Generative AI Tools Are Permitted for Certain Parts of This Assessment.	
Project Inspection	Generative AI Tools Are Permitted for Certain Parts of This Assessment.	
Project Report	Generative AI Tools Are Permitted for Certain Parts of This Assessment.	

Generative AI Tools Are Permitted for Certain Parts of This Assessment (Explained)

In the LT Spice Assignment, you are permitted to use generative AI for the purpose of design concept suggestions, proof reading and editing the document, and for summarising knowledge. No other use of generative AI is permitted.

In the Project Inspection assessment, you are permitted to use generative AI for the purpose of design and build concept suggestions to help with progress in the construction and testing of your project hardware. No other use of generative AI is permitted, and is not permitted to be used during the Project Inspection event.

In the Project Report assessment, you are permitted to use generative AI for the purpose of proof reading and editing the document, and for summarising knowledge. No other use of generative AI is permitted beyond what was permitted for the Project Inspection preparation and LTSpice Assignment.

To assist with maintaining academic integrity, you must appropriately acknowledge any use of generative AI in your work. You must include a Statement of AI use (if no AI tool has been used, then this must also be stated) and a listing of all prompts provided to the AI tool, clearly indicating which AI tools were used and how they contributed to each separate assessment identified (namely LT Spice Assignment, Project Inspection, and Project Report).

Self and Peer Review: Each group member will give each of the team members (**including themselves**) a mark out of 10 on how they contributed to the overall group effort. Each mark must be justified with comments. Each group member will submit their assessment on Learn, and this will be used to scale 30% of the group project mark. **Note: Equitable project work is expected (course outline identifies a nominal 36 hours per group member as reasonable). Issues with group work equity must be raised with the project coordinator lecturer at least two weeks prior to the Lab Review date.**

Race day (Optional: in study week - weather dependent)

Completion minor prize - car completes one of the race heats without assistance.

Runner-up certificate - car gains 4th, or 5th place in final race.

Top place certificate - car gains 1st or 2nd or 3rd place in final race.